

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
FACULTY OF ENGINEERING AND TECHNOLOGY
SCHOOL OF COMPUTING



SRM Institute of Science and Technology
School of Computing



COURSE PLAN
21CSE463T-CLOUD STRATEGY PLANNING AND
MANAGEMENT
JULY - DEC 2026

Revision History:

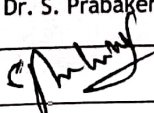
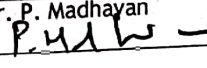
Date	Version	Modification done	Modified by	Reviewed by	Authorized by
20.7.2026	1.0	Initial Release	Dr. S. Prabakeran	Dr. P. Madhavan	Dr. P. Madhavan
					

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1.0 General Details

Course Code: 21CSE463T

Course Title: Cloud Strategy Planning and Management

Semester: VII

Course Time: JULY - DEC 20Slot: C

Day	Batch			
	Batch 1		Batch 2	
	Hour	Timing	Hour	Timing
Day order 1				
Day order 2				
Day order 3	1,2	8:00am - 9:40am		
Day order 4	10	4.00 pm - 4.50 pm (Optional)		
Day order 5	3	9.45 AM - 10.35 AM		

Location: LH1313 - TECH PARK II & LH1314 - TECH PARK II

Name of the Faculty Handling the course

- 1) Dr. S. Prabakeran, Associate Professor, NWC
- 2) Dr Gouthaman P, Assistant Professor, NWC

Student Strength

Batch 1 (Both Faculty)

Section1 (Dr. S. Prabakeran) : Count = 61

Section 2 (Dr Gouthaman P) : Count = 51

2.0 Reference Books

1. Dimitris N. Chorafas: Cloud Computing Strategies, CRC Press, 2011.
2. Arnold J Cummins, "Easiest Ever Guide to Strategic IT Planning"<http://strategicitplanningguide.com/>.
3. David S. Linthicum, "Cloud Computing and SOA Convergence in Your Enterprise", Addison Wesley [ISBN: 0136009220],2009.
4. Charles Babcock, "Management Strategies for the Cloud Revolution", 1st Ed., Tata McGraw/Hill [ISBN: 0071740759],2010.

3.0 Prerequisites

Nil.

4.0 Instructional Objectives

1. Demonstrate how cloud computing enables IT Transformation and business value in an organization
2. Examine the role that cloud computing can play in the business process
3. Appraise how the incorporation of cloud computing in an IT strategy can deliver on strategic business objectives
4. Criticize how cloud computing and Service Oriented Architecture (SOA) can deliver business agility
5. Implement IT governance to manage business realization from cloud IT services

5.0 Overall Assessment Plan

#	Component	Type	Marks
1	FT-1	Experiential Learning vibe coding	5
2	FT- II	Written Test	15
3	FT- III	Written Test	15
4	FT -IV	Mini Project with any Cloud service providers	15
5	LLT-1	Global Certification - Azure	10
Total Marks			60

6.0 Tentative Test Schedule

#	Tentative date	Test	Marks	Portion	Duration
1	11.8.26	FT-1	5	Unit 1	50 Minutes
2	31.8.26	FT- II	15	Unit 1 and Unit 2	100 minutes
3	16.10.26	FT- III	15	Unit 3	100 minutes
4	30.10.26	FT-IV	15	All 5 units	100 minutes
5.	21.10.26	LLT-I	10	AZ-104 — Microsoft Azure Administrator	

7.0 Detailed Test Plan

Test	Tentative Date	Type	Marks	Mode
FT-I	11.8.26	Activity	Experiential Learning vibe coding (Unit 1)	Physical
FT-II	31.8.26	Written Test	Exam Pattern: (Unit 2 & 3) MCQs –14 (14*1=14) (14 marks) Concept Understanding Questions - 4 4*4=16 (16 marks) Scenario based Questions – 2 (2*10=20) (15 marks)	Physical Exam
FT-III	16.10.26	Written Test	Exam Pattern: (Unit 4 & 5) MCQs –14 (14*1=14) (14 marks) Concept Understanding Questions - 4 4*4=16 (16 marks) Scenario based Questions – 2 (2*10=20) (15 marks)	Physical Exam
FT-IV	30.10.26	Written Test	Mini Project – (10 + 5 =15 Marks) Interim Review will be conducted for 5 Marks	Physical Evaluation

LLT1	21.10.26	Global Certification: AZ-104 — Microsoft Azure Administrator	Total :10 Marks AZ-104 — Microsoft Azure Administrator	Global Certification
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Detailed Session Plan

Sl. No	Topics to be covered	Hours	Teaching Method	Testing Method
UNIT-1				
1	The four pillars of cloud computing, Cloud applications and Platforms	1	PPT, BB	Illustration using example
2	Providing the cloud infrastructure, Cloud computing, Spectral efficiency, Sensors and perspiration	1	PPT, BB	Illustration using example
3	Strategic inflection points in information Technology, Cloud computing and its slogans	1	PPT, BB	Illustration using example
4	User centered solution and cloud computing, For cloud vendors inflection point Is risk and opportunity	1	PPT, BB	Illustration using example
5	Potential customers of cloud technology	1	PPT, BB	Illustration using example
6	The cloud interests Small and Medium enterprises	1	PPT, BB	Illustration using example
7	Virtual companies and the cloud, Virtual networked objects	1	PPT, BB	Illustration using example, Group Discussion

8	Consumer technologies and the cloud	1	PPT, BB	Illustration using example
9	Social networks and multimedia messaging	1	PPT, BB	Illustration using example

UNIT-2				
10	Moving to a cloud architecture and strategy to achieve business value	1	PPT, BB	Illustration using example
11	BPM, IS, Porter's Value chain model and BPR as a means of delivering business value	1	PPT, BB	Illustration using example
12	Developing Business Strategy: Investigate business strategy models to gain competitive advantage for organizations, SWOT/PEST, Economies of scale, Porter's 3 Strategies and 5 Competitive Forces, D'Aveni's hyper competition models	1	PPT, BB	Illustration using example
13	Emphasize the roles of the strategic IS/IT leaders such as Chief Information Officer (CIO)	1	PPT, BB	Illustration using example
14	The Chief Technology Officer (CTO) in planning and managing IT Strategic development in the organization.	1	PPT, BB	Illustration using example, Group Discussion
15	Budgeting for cloud computing	1	PPT, BB	Illustration using example, Group Discussion
16	Service level agreements	1	PPT, BB	Illustration using example
17	Outsourcing, Infrastructural inter dependencies, and the cloud	1	PPT, BB	Illustration using example, Group Discussion
18	Human resources at the CIO level, The transition from legacy to competitive system	1	PPT, BB	Illustration using example

UNIT- 3

19	Develop an IT strategy to deliver on strategic business objectives in the business strategy	1	PPT, BB	Illustration using example
20	IT Project planning in the areas of ITaaS is essential in delivering a successful strategic IT Plan	1	PPT, BB	Illustration using example
21	IT Project planning in the areas of SaaS essential in delivering a successful strategic IT Plan, IT Project planning in the areas of PaaS essential in delivering a successful strategic IT Plan	1	PPT, BB	Illustration using example, Group Discussion
22	IT Project planning in the areas of IaaS essential in delivering a successful strategic IT Plan	1	PPT, BB	Illustration using example
23	Searching for an open architecture	1	PPT, BB	Illustration using example
24	Infrastructure as a Utility	1	PPT, BB	Illustration using example
25	Cloud System Architecture and its primitives	1	PPT, BB	Illustration using example
26	The User Organizations Business Architecture	1	PPT, BB	Illustration using example
27	Financial Services Applications Architecture	1	PPT, BB	Illustration using example

UNIT-4

28	Shared services delivered by a Service Oriented Architecture (SOA) in a Private or Public Cloud	1	PPT, BB	Illustration using example
29	Services, Databases and Applications on demand	1	PPT, BB	Illustration using example, Group Discussions

30	The effect on Enterprise Architecture and its traditional frameworks such as Zachman), The Open Group Architecture Framework (TOGAF).	1	PPT, BB	Illustration using example
31	Customer Relationship Management	1	PPT, BB	Illustration using example
32	Enterprise Resource Planning	1	PPT, BB	Illustration using example
33	Just-in-Time Inventories	1	PPT, BB	Illustration using example
34	Machine-to-Machine and RFID Communications	1	PPT, BB	Illustration using example
35	Challenges Presented by Organization	1	PPT, BB	Illustration using example
36	Challenges Presented by Commercial vision	1	PPT, BB	Illustration using example
UNIT 5				
37	Benefit Realization and it Governance	1	PPT, BB	Illustration using example
38	Managing resources (people, process, technology), to realize benefit from Private/Public Cloud IT services	1	PPT, BB	Illustration using example
39	Gartner's 5 pillars of benefit realization, IT governance as a service in measuring the delivery of IT Strategy from Cloud IT Services using Sarbannes Oxley (CobiT) and other commonly-used approaches	1	PPT, BB	Illustration using example
40	High Technology for private banking and Asset Management	1	PPT, BB	Illustration using example
41	Cloud Software for Private Banking	1	PPT, BB	Illustration using example
42	Leadership Is based on Fundamentals	1	PPT, BB	Illustration using example
43	Cloud Software For Asset Management	1	PPT, BB	Illustration using example

44	Cloud Technology can Improve Fund management	1	PPT, BB	Illustration using example
45	Criteria of Success in Asset Management Technology	1	PPT, BB	Illustration using example

Experiential activity (Rubrics)

S.No.	Evaluation Criteria	Excellent (Full Marks)	Good	Satisfactory	Needs Improvement	Marks
1	Problem Identification & Requirement Analysis	Clearly identifies a real-world problem, stakeholders, functional & non-functional requirements with strong justification.	Problem and requirements are mostly clear.	Basic problem statement with limited analysis.	Problem is vague or poorly defined.	10
2	Alignment with Course Syllabus	Demonstrates excellent application of the assigned cloud strategy concepts from the syllabus with clear mapping.	Most syllabus concepts are appropriately incorporated.	Limited mapping to syllabus topics.	Poor or no connection to syllabus.	10
3	Solution Design & Cloud Architecture	Well-designed architecture with proper cloud deployment model, scalability, security, architecture diagrams, and justification of services.	Good architecture with minor omissions.	Basic architecture without sufficient justification.	Incomplete or incorrect architecture.	15
4	Use of Vibe Coding Tools	Effectively uses AI-assisted coding tools (e.g., Cursor AI, GitHub	Good use of AI tools with minor dependence.	AI tools used but limited understanding/customization.	Minimal or ineffective use of AI tools.	10

SRM S.No.	Institute of Science and Technology, Kattankulathur Evaluation Criteria	Excellent (Full Marks)	Good	Satisfactory	Needs Improvement	Marks
		Copilot, Bolt.new, Lovable, Replit AI) with appropriate prompts and demonstrates understanding of generated code.				
5	Implementation & Functional Completeness	All planned features are implemented correctly; application works reliably without major defects.	Most features implemented with minor issues.	Core features implemented; several functionalities missing.	Major functionalities incomplete or non-functional.	20
6	Cloud Service Integration	Appropriate integration of cloud services (authentication, database, storage, APIs, hosting, serverless functions, etc.) with proper configuration.	Good cloud integration with minor issues.	Basic cloud integration.	Very limited or no cloud integration.	10
7	User Interface & User Experience (UI/UX)	Professional interface, responsive design, intuitive navigation, accessibility, and excellent user experience.	Attractive interface with minor usability issues.	Functional but basic interface.	Poor interface and difficult navigation.	5
8	Innovation & Creativity	Solution introduces innovative features, unique ideas, or intelligent automation beyond minimum requirements.	Some creative enhancements.	Limited innovation.	No innovation beyond basic implementation.	5
9	Testing & Quality Assurance	Comprehensive testing, debugging,	Good testing with minor gaps.	Basic functional testing only.	Little or no evidence of testing.	5

SRM S.No.	Institute of Science and Technology, Kattankulathur Evaluation Criteria	Excellent (Full Marks)	Good	Satisfactory	Needs Improvement	Marks
		validation, edge-case handling, and performance evaluation documented.				
10	Documentation Quality	Comprehensive documentation including problem statement, requirements, architecture, screenshots, AI prompts, code explanation, testing, deployment, and references.	Good documentation with minor omissions.	Basic documentation.	Poor or incomplete documentation.	5
11	Presentation & Demonstration	Professional presentation, effective live demonstration, confident explanation, and excellent responses to questions.	Good presentation with minor issues.	Average presentation.	Poor presentation or inability to explain work.	5
	TOTAL					100

Global Certification Details

Course Title: Microsoft Azure Administrator

Certification: Microsoft AZ-104 – Associate Level

Duration: 4 Days

Mode: Offline

The course provides practical training in managing Microsoft Azure services. It covers:

- Identity and governance using Microsoft Entra ID
- Azure storage accounts, Blob Storage and Azure Files
- Virtual machines, containers, App Service and Docker
- Virtual networks, subnets, peering, NSGs and load balancing
- Azure monitoring, alerts, backup and disaster recovery
- Guided project, hands-on labs and AZ-104 exam preparation

Link : <https://learn.microsoft.com/en-us/credentials/certifications/azure-administrator/?practice-assessment-type=certification>

Mini Project Evaluation Rubric (100 Marks)

S.No.	Evaluation Criteria	Description	Marks
1	Problem Analysis & Solution Planning	Understanding the problem, identifying requirements, and proposing an appropriate cloud-based solution.	10
2	Application Design & Architecture	System workflow, architecture diagram, cloud deployment model, scalability, and technical design.	15
3	Functional Implementation	Completeness of features, correctness of implementation, and overall functionality of the application.	25
4	Cloud Integration	Effective use of cloud services such as authentication, storage, databases, APIs, hosting, or serverless computing.	10
5	Use of Vibe Coding / AI Tools	Effective use of AI-assisted coding tools, prompt engineering, code customization, and understanding of generated code.	10
6	Innovation & Creativity	Originality, problem-solving approach, additional features, and creative implementation.	10
7	User Interface & User Experience (UI/UX)	Attractive interface, ease of use, responsiveness, and user-friendly navigation.	5
8	Testing & Debugging	Testing of application features, handling errors, debugging, and application reliability.	5
9	Project Demonstration	Live demonstration of the application, explanation of features, workflow, cloud services used, and overall presentation.	10
	TOTAL		100

Interim Review Instructions

The **Interim Review** will be conducted during the **last week of August**.

The interim review carries **5 marks out of the total 15 marks** allotted for the mini project.

Students must present the following during the review:

1. **Industry Use Case aligned with an SDG Goal**

Select a real-world industry problem and clearly map it to one or more **United Nations Sustainable Development Goals (SDGs)**.

2. **Project Objective and Expected Outcome**

Clearly explain:

- What problem the project addresses
- The main objective
- The expected result or impact

3. **Architectural Design**

Present the proposed cloud architecture and list at least **4–5 major features or modules** of the application.

4. **Technology Stack**

Mention the selected cloud platform and development technologies:

- AWS, Azure, or GCP
- Frontend
- Backend
- Database
- Frameworks or tools used

5. **Cloud Services Used**

The project must use a minimum of **five cloud services**.

Basic services such as **EC2 and S3 alone will not be considered sufficient**. Students are encouraged to use advanced services such as:

- Serverless computing
- Managed databases
- AI/ML services
- Monitoring and logging
- API management
- Identity services
- Messaging and event services
- Analytics services

6. **Completion of One Module**

At least **one project module must be completed and demonstrated** during the interim review.

Students must explain all the above points and provide a working demo of the completed module. The review will assess project clarity, technical planning, cloud-service selection, architecture, and implementation progress.